

TwinkleBands: Wearable Technology Supporting Movement Imitation and Coordination in Fast-Paced Fitness Training

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ABSTRACT

Fast-paced fitness training (e.g. aerobics, Bodycombat, step workouts) is one of the most popular activities at fitness clubs worldwide because it is beneficial for physical health and it can enhance participants' motivation and engagement of fitness training. Yet, their fast pace and required coordination make it difficult for some participants, especially for beginners and those with coordination problems, to follow the class. Here we present the design, implementation and qualitative user evaluation of TwinkleBands, a piece of wearable technology that provides real-time support with trainees' movement and coordination learning by providing discriminative visual cues on the extremities. We show that TwinkleBands helps movement imitation and coordination in several ways. Based on this, we discuss key design takeaways for future technology design to support movement teaching and learning in fast-paced activities.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction devices.

KEYWORDS

Fast-paced training, Movement learning, Wearable devices

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1 INTRODUCTION

Teaching and learning movement is complicated. In their everyday life, people engage in different movement practices (sports, fitness, somatic practices) that feature a strong pedagogical component towards the development of particularly relevant bodily skills and embodied knowledge [34]. The reasons to partake in those are varied, and range from mental and physical wellbeing reasons, to social belonging and community building [34].

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In the third wave of HCI, the focus on leisure and movement-based interaction has put these practices on the spot. A plethora of technology solutions to support movement teaching and learning have emerged, showing the technology's potential to help people bridge common challenges in movement learning, such as developing nuanced body awareness, support execution of movements, and identify and correct errors. In sports/fitness HCI, advances in computational capacities, and a strong technology-driven field development [24] have favored strong data-driven design approaches, that rely on movement detection and processing [39], such as famous commercial solutions for home training [42], and exergaming [22]. However, researchers are still exploring how to develop and integrate technology to support complex movement practices that are executed at high-speed and require fast and accurate coordination [28, 35, 37], such as dancing-inspired workout and pre-choreographed fitness practices (e.g. Zumba [5], Les Mills Bodycombat [2]).

These practices are often social and instructed: a teacher is showing in real time how to perform the movements, which a group of trainees have to imitate. In these, the training session is often continuous: it is not allowed to be paused, and cues and feedback are often not tailored to particular individuals in the same class. These types of practices pose particular challenges for technology inclusion real-time: their fast pace and full-body, complex movement coordination demand high attention and focus on the trainee's side, and technology inclusion often risks becoming obtrusive, or distracting [35, 39]. Due to these very challenges, executing and coordinating movements is challenging itself, and people could benefit from technological interventions to help them in movement learning [28, 35, 37].

In this work, we explore how to design technology to support movement teaching and learning in these fast-paced training activities. We focused on a choreographed and instructed fitness practice (Les Mills Bodycombat), as a representative of our targeted practices. Following a Research through Design approach [25], we centered on exploring the common challenges in movement learning in those settings. These relate to trainees' following the instructor's demonstration and executing bodily movements correctly, especially for beginners who step into a class and those with coordination problems. Meanwhile, instructors also need a supportive tool to teach bodily movements and correct trainees from errors, and it should not affect their existing demonstration and coaching in the class.

Through a series of participative, embodied design [40] workshops, we developed a simple wearable system, TwinkleBands. This consists of four wearable bracelets (2 for wrists and 2 for ankles) that both instructor and trainees wear in their extremities. They

provide color-based discriminative visual cues that visually augment the body laterals (right/left) and focus attention on relevant joints and movement combinations.

We integrated the use of TwinkleBands in a short Bodycombat session (20 minutes), and conducted a preliminary user test with five trainees as participants. Results show that the inclusion of TwinkleBands supported trainees in understanding, following and imitating the instructor's demonstrations, and helped them in coordinating complex, full-body movements according to the choreography. It also showed the potential to help trainees to autonomously identify and correct errors. Based on these results, we discuss preliminary design takeaways and sketch future work to keep exploring how to develop and integrate technology in fast-paced movement learning activities.

Our contributions are three: (1) an investigation of common problems in fast-paced fitness training; (2) a semi-functional prototype generated from design and research, and results from evaluations in a real training scenario; and (3) preliminary inspirations for design. These contribute to the growing body of work in HCI investigating how to design technology integration in complex and fast-paced training activities (e.g. [19, 20, 29, 35, 37]).

2 BACKGROUND

Choreographed fitness training is an important form of training that features in the everyday life of people [2, 18]. They are beneficial for trainees' cardiovascular system, neuromuscular function, flexibility and coordination [5, 30]. They can also improve trainees' motivation by engaging them with music and other trainees in the class [12, 15]. In this type of physical training, one instructor demonstrates bodily movements physically and gives oral cues simultaneously in front of a group of trainees collocated in the same room. Trainees conduct training by imitating the instructor's movements, and improve by practicing. The choreography is pre-composed with music, and its complex and fast-changing move combinations (move combos) demand fast and high-level four-limb coordination.

In fast-paced fitness training, trainees who are not familiar with movements need to concentrate on the instructor's demonstration and listen to the oral cues. As an example of these types of practices, we focus on Les Mills Bodycombat, a choreographed fitness practice to train endurance, fitness and strength. It mixes a variety of combat moves (e.g. punches, kicks) from a variety of martial practices (e.g. kickboxing, karate) that are synchronized to the beats of a song. A normal class lasts around 1h and features 8-10 songs, each with a distinct choreography.

Research regarding group fitness exercises in HCI has paid attention to its social aspects and to real-time data monitoring [6, 23]. In relation to movement learning itself, prior research in the domain has shown that joints (e.g., knee, wrist, shoulder...) can be used as essential hints for trainees to understand bodily movement, e.g.: YouMove is a movement training system by using a half-silvered mirror with graphic overlays for guidance and feedback [1]. In this study, important joints are highlighted as movement parameters that are used in the system to provide guidance and feedback. Another movement guidance technology called AR-arm gives egocentric hints to guide the participant's arm movement by implementing

a video see-through head-mounted display in the first-person perspective [11]. Considering this, body joints can be vital visual cues and references for participants to perceive, pay attention to, and synchronize their own bodily movements to these hints, and finally achieve the correct movements and four-limb coordination. We have taken inspiration from these studies to develop TwinkleBands.

2.1 Visual Cues and Augmented Feedback to Support Movement Learning

Our wearable solution offers color-based visual discriminative cues. This modality can convey rich and nuanced information [27, 38] in a way that people can easily make sense of [8, 27]. Previous studies have shown that visual feedback can support people's execution on body alignment [36] and performing movements that require accurate technique [32]. Visual cues and feedback can also help participants to learn techniques in physical exercises [25]. In fast-paced fitness activities, participants who are not familiar with movements need to concentrate on the instructor's demonstration and listen to the oral cues first of all. Considering the fast-paced scenario, visual cues can be less intrusive compared with other sensorial-augmented cues such as haptic and auditory feedback [7]. Moreover, in complex move combos asking for high-level four-limb coordination, the visual color can play a role as a discriminative cue in people's learning of different mappings according to [4]. In other technology supporting physical training, color cues, as a type of flexible information coding, can be a display mechanism to indicate communicate multiple types of feedback by changing their patterns and lightness [3, 7, 41]. Ballet Hero, an e-textile garment designed for adult learners to mimic instructor's movement, also used color-based patterns as visual feedback [10]. This design is inspiring, but it did not attempt to resolve coordination problems, as it focused on detecting the full range of motion of a dance instructor instead of helping learners to identify their body sides. Harrison et al.'s research [13] also inspired us to attach visual stimuli onto people's various locations of body. These studies serve to underpin our design decision of using visual cues to augment bodily movements and using colors to differentiate cues of two body sides (right/left).

3 METHODS

Our work was conducted in four consecutive phases. We briefly discuss them and how they influenced each other.

3.1 Phase 1: Understanding Problems in Movement Imitation and Coordination in Fast-Paced Activities

First, we engaged in a qualitative study to understand prevalent issues in movement teaching and learning in fast-paced environments, where imitation and coordination is key. We conducted semi-structured interviews with 9 practitioners of Bodycombat, recruited from our pool of contacts: 5 trainees (T=5: 2 beginners; 2 intermediates and 1 advanced) who train Bodycombat on a regular basis (≥ 60 min per week); and 4 certified instructors (I=4). Each interview lasted around 45 minutes and was audio recorded. Questions focused on identifying and understanding teaching and learning issues, and what strategies they used to recover from errors.

Interviews were transcribed using an inductive thematic analysis approach [9]. The first author conducted the first round of open coding and engaged in discussions with the second to perform two rounds of axial coding. Here we summarize the main insights that shaped the subsequent phases.

3.1.1 Relevant Insights. The analysis revealed that the most common problematics of participants related to: (1) **Difficulties in following the instructor's movement demonstration in real time.** In pre-choreographed training activities such as Bodycombat, the instructor's and the trainees' movements occur at the same time, and the former cannot pause to show slow motions or decompose the move combos. This makes it difficult for participants who are not very familiar with choreography to imitate and coordinate the movements, as captured by T2's quote: "I could not follow the movement of the instructor. . . I cannot memorize them either." This was echoed by all the instructors, as captured by I3's quote: "The challenge is that the changes [of moves] are fast. If you are a person who is not familiar with choreography or uncoordinated body, it's difficult." (2) **Difficulties in coordination and understanding due to the face-to-face setting.** In the setting of a training class, participants mirror the instructor's demonstration, e.g.: the movements that the instructor executes on their right arms/legs are imitated by the trainees on their left. This is complex for some trainees, and is accentuated in full-body workouts like Bodycombat require complex four-limb coordination in multiple movement planes. Understand what limb to move, and how to coordinate it with the others based on the mirrored view of the instructor is challenging, as captured by T4's quote: "It's hard to see when [instructors] are moving the right arm and left leg as a combination". This mirror setting also makes oral directional cues given by instructors (e.g. right/left) confusing: "when hearing left and right, I need to flip (instructor's) direction to mine". (3) **Difficulties in movement correction.** As the fast-paced, choreographed training setting offers little space for instructors to correct individual errors, trainees felt it was hard to autonomously identify errors or correct them. T1: "I mimic the instructor, but I don't know whether I moved [my body] to that position [correctly]." Instructors expressed this was a central concern when trying to correct people in real time: "[I find] people are struggling in a specific thing. But you (instructor) have to carry on, it's not possible you pause the class to go and pay attention to each individual one" (I2).

Based on these insights, in Phase 2 we engaged in embodied design workshops to explore how a technological intervention could help address or mitigate some of these issues.

3.2 Phase 2: Two Embodied Design Workshops

Influenced by participatory design [33] approaches, we organized two workshops around Bodycombat with instructors (W1 W2 each one week apart), trainees at different levels and interaction designers. W1(n=5) had 2 instructors and 3 trainees; and one of the instructors played the role of a designer. W2 (n=3) had 2 trainees from W1, and one instructor who was also the designer of W1. We employed embodied design methods for movement-based design [21], which foreground bodily engagement with bodily-technical-social design materials [14], to design compelling body experiences

supported by technology and integrate such technology into complex movement practices.

W1 focused on ideation: producing different ideas for technological solutions. The 1st author introduced participants to the problems identified in the prior phase. To sensitize and prepare participants for ideation, the 1st author facilitated a short 15' Bodycombat class in which trainees were asked to reflect on the identified problems as they performed the movements. After the class, a discussion ensued to further discuss the lived experience of the participants in relation to the problems, which further confirmed them. Based on this sensitizing, the 1st author facilitated a situated brainstorming session in which technological solutions were proposed, discussed, discarded and refined. Two (rather vague) design ideas were found interesting by all participants of W1: (1) wearable LEDs to augment visual cues of the instructor's bodily movement to indicate the engaging limb(s); (2) wearable device/gloves with haptic feedback on participants' body if they perform incorrect movements.

W2 centered on developing and exploring these ideas further. Various materials (e.g. colorful papers, tapes, ribbons) were brought in to create low-fi prototypes. Through bodystorming techniques inspired in [31], we explored their use in different moves and combos, and jointly discussed how successful they were in supporting movement imitation and coordination. The wearables' visual solution was the preferred one: it was less intrusive because it was easy to understand and can be shared within a group of participants. The haptic ideation in the fast-paced environment was described as "distractive" and generated too much mental stress. The ideation of interface drew trainees' attention from instructor to a screen and was difficult to be integrated into the existing fast-paced environment. With the resulting idea from W2, we proceed to phase 3, in which we designed and iterated our technology: TwinkleBands.

3.3 Phase 3: Designing TwinkleBands

The TwinkleBands system (see Figure 1) is made up of four wearable devices (2 for wrists and 2 for ankles) which are not bulky and durable in fast-pace workouts. The compounds of each device include: an elastic strap; a rechargeable battery mounted with a hidden rubber band inside; a Neopixel LED strip; an Adafruit trinket mini board; and a slide switch. The hook & loop of the elastic strap makes it can be fastened and fit easily on the wrist and ankle. A 12-LED Neopixel strip is sewed on the strap. To make the light attractive but not intrusive, we coded a delay between each LED. This makes the illuminant loop around the strap to convey the visual cues in a mild and comfortable way. We selected green and red colors to differentiate the left and right body sides. This design was arrived at through an iterative design process in which we invited for short testing sessions including one instructor, one trainee and one designer. We evaluated different body locations, different methods of attaching them to the body, and different visual cues (frequency; color; and brightness). After several trials and discussion, we found that bands or bracelets worn on the instructor's body can be detected by participants unhindered when stances and directions were changed in the demonstration.

The instructor can manually switch on/off each device in different combinations (Figure 2). This is a not fully functional implementation which needs the instructor to pause and manipulate but



Figure 1: The design prototypes and their implementation on body



Figure 2: Color-based cues in different move combos: a, b, c, d are in front stance (all switched on); e, f, d are in combat stance and moves on one body side only (one-side switched on); a trainee wear the low-fi prototype with the same color in opposite body side(h).

enabled us to conduct early user evaluations. We did not prescribe which side is green or red. The instructor randomly chose the same color to wear on one side of the body, and the participants wear the same color on their opposite body side. This was envisioned to help trainees imitate and coordinate limbs in a mirror-flip way (Figure 2).

3.4 Phase 4: Evaluating TwinkleBands

Finally, we performed a first round of evaluations in a semi-real setting of fast-paced training. We invited 5 participants as trainees ($P=5$) to try our prototype in a 20-minute Bodycombat class led by a certified instructor. To comply with safety measures due to the Covid-19 pandemic, trainees were divided into three sessions (2 sessions with 2 trainees respectively, and 1 trainee was tested individually – all with the same instructor). In each session, trainees were asked to do a 5-minute warm-up session without any device to prepare them for the upcoming complex movements. Then the instructor switched on the TwinkleBands and participants wore the strips with the same colors but on opposite body sides (see Figure 2). After wearing the device, they were asked to practice another 15-minute session of more complex and combinations (e.g. stepping jab, side kick, roundhouse kick, split jack...) in different stances (front stance, right/left combat stance). In the second session, the instructor needed to pause her demonstration twice to switch on/off the device when the stance changed. However, this pause did not affect or slow down the flow of movements. Because pauses were within the time gap designed in choreograph for stance change. The evaluation was recorded by a camera and a semi-structured post-interview was conducted. The data of in-depth interview were transcribed and analyzed thematically to understand how participants used TwinkleBands and their feedback. We also used video

analysis [16] as supportive evidence to retrieve and confirm the essential details which participants described or mentioned in interviews, especially for their self-correction.

4 RESULTS

Even though TwinkleBands is a not fully functional implementation, requiring the instructor's manual operation, the results from our analysis show that TwinkleBands positively impacted the trainees' understanding and execution of movements and complex limb coordination in fast-pace fitness training. We briefly describe the main effects below:

Enhancing instructor's demonstration with discriminative visual cues. With TwinkleBands, two illuminant colors (green and red) in different body sides (right and left) provide trainees with discriminative visual cues that augment the lateral sides of the body. This offered an easy-to-understand way for trainees to match their limbs' movements to that of the instructor. As visual cues, particularly colors, are easy to understand for people [8, 27], it enabled trainees to make sense of the mirror situation in fast-paced and complex move combos. With TwinkleBands, they could easily identify body segments with according colors on their main joints: "with the TwinkleBands...the information is clearer, and I know which hand/leg the instructor is using" (P2). TwinkleBands visually strengthened directional oral cues, which are important during instructor's demonstrations but sometimes difficult to grasp. E.g. cues such as "release your heel and rotate the torso". As captured by P1: "with [TwinkleBands], I see you are releasing the back heel because the light is on your ankle maybe and twisting the lower body on uppercuts, [...] which I have never noticed before". Compared with the setting without TwinkleBands, the visual cues are more intuitive for them to detect: "[without TwinkleBands] each



Figure 3: a) P5 executes the incorrect knee and detects the different colors. b) P5 pauses and checks the TwinkleBands on her own body and c) gets the right coordination. d) P2 executes the opposite arm in a split squat. e) without a pause, P2 corrects the move.

time when I hear the right-left cues you give; I need to flip your images and make it as my [left/right] ... the cues of different colors of the right and left body are more intuitive than oral cues" (P2).

Supporting imitation and coordination. With TwinkleBands, fast-paced complex move combos are augmented with four luminous points on the instructor's extremities, to which trainees could orient as references for understanding the movement. By matching those points to those in their own body, all trainees felt that their understanding and awareness of their own performance was enhanced: "The lights make it [moves] easier to understand" (P5). TwinkleBands was also experienced in helping in imitation. By establishing a connection between color-based visual cues with the according limbs of their own body, participants matched their own colors with instructor's. This visual-kinesthetic connection helps them synchronize the instructor's demonstration to their own body in fast-paced movements. "I look at your colors first, then I look at mine. When training, I match your colors with mine" (P3); "[it] shortens the process of transferring the cue such as right and left into my own body" (P2).

Emerging imitation/coordination strategies. In the interviews, several trainees discussed emerging imitation/coordination strategies enabled by the inclusion of TwinkleBands. Some established a visual-kinesthetic connection by focusing on particular body parts to imitate: "I pay more attention to the light on your left leg; ...so I take your left leg [with LEDs] as an anchor. If your left leg is stepping forward, I know I need to step my right leg out ... and to navigate the left three ones" (P1). Others mentioned that they used TwinkleBands to memorize the movement combos as color sequences: "I transfer the moves to the [colors of the] light. In previous training, I memorized and did it (the move combo) as punch (right) punch (right), punch (front) [...] Now [with colors] I know the combination is red, red, green" (P2). Finally, some could easily establish visual-kinaesthetic connections early in the class, so they did not check their own color(s) that much in fast moves: "I have this [color-body] connection when wearing the light [...] I knew this part (gesticulates her right) is green and the other is red. Then I don't need to check my colors all the time" (P5). These different strategies show that the open-ended design of TwinkleBands enabled trainees to come up with, and tailor, relevant strategies that helped in coordination and imitation.

Towards autonomous movement correction. As TwinkleBands supported trainees in autonomously identify and correct movement errors (see Figure 3). They would detect errors by using the instructor's TwinkleBands as checking points to monitor if their

colors match with instructors. This was found to amplify errors: "the lights make my mistake so obvious. [I was moving to] the opposite direction!" (P4). Some trainees had to pause to correct their performance, while others were able to correct without pausing or checking their own colors, which was attributed by someone having more experience with the technology: "I think I am an advanced user of TwinkleBands because I have [now] used it for a while. I can connect your light to my body and don't need to check my color now."

5 DISCUSSION AND FUTURE WORK

Through the design and evaluation of TwinkleBands, initial design takeaways emerged that hold the potential to be relevant when designing technology for fast-paced movement teaching and learning practices.

Simplicity. TwinkleBands is a very simple solution capitalizing on discriminative visual cues to support movement imitation and coordination. Users can understand and execute bodily movement and complex move combos in a fast-paced scenario, and can even recover from errors after establishing a solid connection between visual mappings and kinesthetic awareness. Its technical and use simplicity enable both trainees and instructors to use them alongside the complex information that they obtain from their bodies, other's bodies, instructions, and so forth. Our initial results seem to indicate that TwinkleBands did not add extra workload to participants, in terms of changing their practice or cognitive load in making sense of the visualizations, but future work is needed. TwinkleBands augments information that already exists in the training (e.g. move combos) in a non-intrusive way, in contrast to other ideas explored in design workshops that were deemed either too aggressive or intrusive for participants in cardio training and physically in aerobic condition.

Visual discriminative cues. TwinkleBands contribute to the use of visual cues in movement education [35, 36]. Its colors are identical and flexible; they can be coded intrusively to convey information as a display mechanism. TwinkleBands shows promise to support persistence of vision [17]; it possible for participants to see the trajectory of instructor's movements. TwinkleBands simple visual design allows people to orient to it without completely switching the focus from other important elements in learning (e.g., posture, power and techniques, instructor's body). Its visual simplicity also avoids common pitfalls of visual augmentations in the domain, such as distraction and undesired body postures and gaze

focus [36, 39]. Further work is required to explore discriminative cues.

Open-ended. TwinkleBands offers an open-ended visualization that amplifies movements, rather than providing information on whereas those movements are right or wrong. This is shared with a growing body of work for movement teaching and learning that employs open-ended output to enable people to construct their own meanings (e.g. [20, 26, 36, 39]). This solution is particularly relevant for fast-paced fitness training, in which all participants with different physical conditions share the same speed, the same movement, the same choreography with only one instructor; and in which it is impossible to pause or tailor the instructions and feedback to particular situations. Open-ended technology offers versatility in that regard: people can come up with their own use strategies and support autonomous corrections.

The design takeaways help shape future research directions on technology design for fast-paced training activities, and we will explore them further. This will entail deploying TwinkleBands in a realistic setting (a full-hour Bodycombat class with multiple trainees) and in other similar practices (e.g. aerobics). This will help us further elucidate the effects of introducing simple discriminative visual cues and derive stronger and more fully fleshed design takeaways in the form of technology design recommendations for fast-paced movement learning practices.

6 CONCLUSION

Our work contributes to the growing body of work on technology design and integration in fast-paced movement learning settings (e.g. [19, 20, 29, 35, 37]) in three ways: (1) bringing forth common problems in movement learning in fast-paced fitness training; (2) TwinkleBands as design particular, and results from its evaluations; and (3) preliminary design takeaways for the domain. These can serve as inspirations for other designers to keep opening up the design space of technology for motor skill development in complex contexts of use.

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